Python & Statistics Challenges

Navigation Product Analytics — 10 Analysis Tasks

Based on tables: users, navigation_sessions, route_options, user_events, experiments, retention_daily

Format: No solutions provided — use pandas, scipy, statsmodels, or scikit-learn

How to use this document

These tasks are meant to be completed against the raw CSVs. Where a task calls for a statistical test, state your significance level up front (conventionally $\alpha = 0.05$), check the relevant assumptions, and report effect size in addition to p-value.

Task 1 — Distribution Analysis

Plot and describe the distribution of recommended_eta_minutes, actual_duration_minutes, and distance_km. Identify skewness and discuss whether a transformation (e.g., log) would be appropriate before modeling.

Task 2 — Correlation Analysis

Compute a correlation matrix across traffic_uncertainty_score, route_complexity, road_quality_score, number_of_turns, reroute_count, and route acceptance (as a binary variable). Discuss which correlations are strongest and whether any are surprising.

Task 3 — Outlier Detection

Use both a statistical method (e.g., IQR or z-score) and a visual method (box plot) to identify outliers in distance_km and actual_duration_minutes. Decide how you would treat them and justify your choice.

Task 4 — Segmentation

Segment users based on behavioral features (acceptance rate, average reroute_count, session frequency, completion rate) using k-means or hierarchical clustering. Profile each resulting segment.

Task 5 — Logistic Regression

Build a logistic regression model predicting route acceptance (accepted vs rejected) from traffic_uncertainty_score, toll_recommended, route_complexity, road_quality_score, and user_type. Interpret the coefficients and discuss model fit.

Task 6 — Hypothesis Testing

Test whether the average reroute_count differs significantly between sessions with High vs Low traffic_level. Choose and justify an appropriate test given the data's distribution.

Task 7 — Confidence Intervals

Construct a 95% confidence interval for the overall recommended-route acceptance rate, and separately for New vs Returning users. Discuss whether the intervals overlap.

Task 8 — A/B Testing

Analyze the experiments.csv data to test whether the Treatment (ETA confidence range) significantly improves route_acceptance versus Control. Report the p-value, effect size, and a practical recommendation, and check at least one guardrail metric.

Task 9 — Feature Importance

Using a tree-based model (e.g., random forest or gradient boosting) predicting navigation_completed, extract and rank feature importances. Compare the ranking to the coefficients from Task 5's logistic regression.

Task 10 — Retention Analysis

Using retention_daily, build 7-day and 30-day retention curves by signup cohort (month of signup). Test whether retention differs significantly between users who experienced at least one abandoned session in their

first week versus those who did not.